Nie D, Wang L, Xiang L, et al. Difficulty-aware attention network with confidence learning for medical image segmentation[C]//Proceedings of the AAAI Conference on Artificial Intelligence. 2019, 33: 1085-1092.

2020.3.12
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论文背景

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出发点

Training of the network is inclined to be dominated by the easy-to-segment samples.

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Deep segmentation networks cannot properly handle the hard-to-segment samples (or regions)

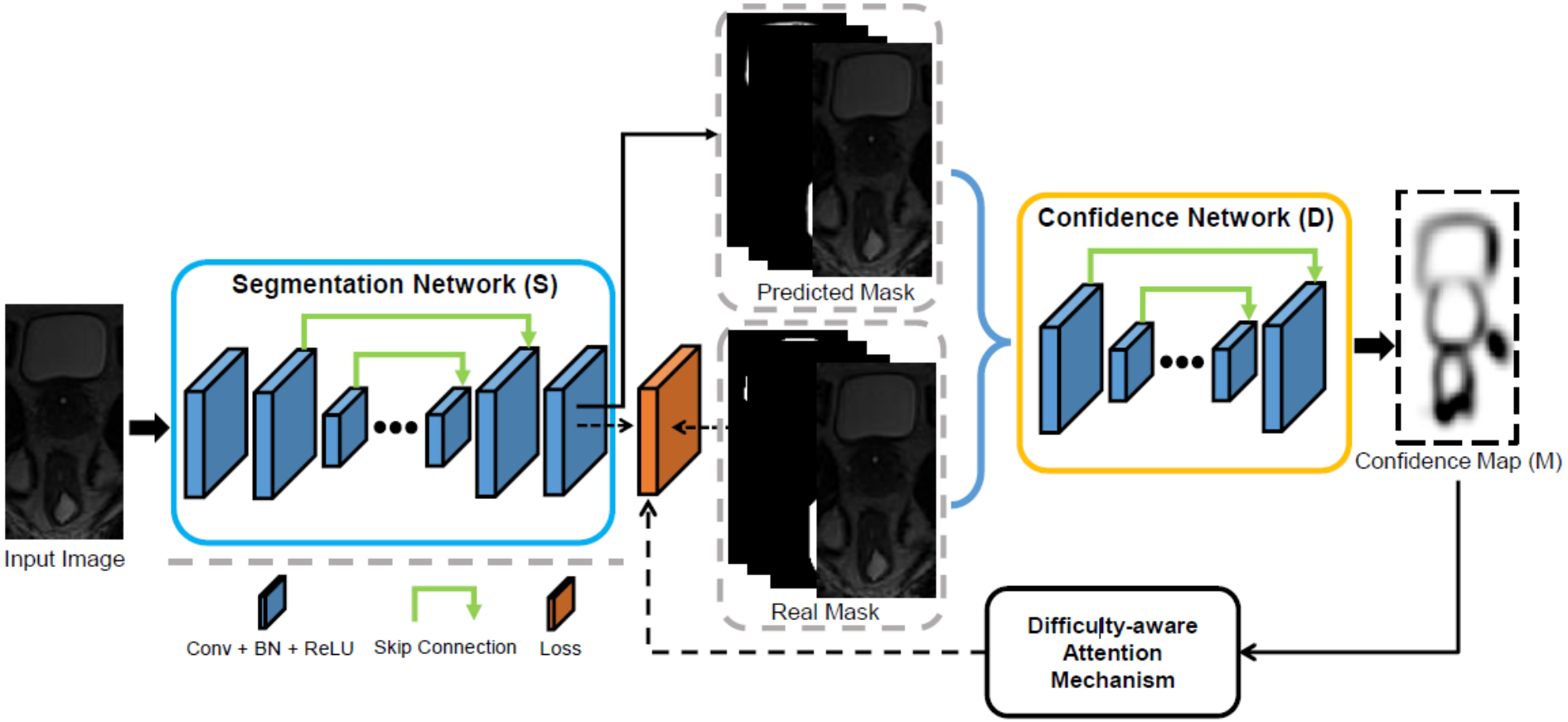

Figure 1: Illustration of the architecture of the proposed framework. This framework consists of a segmentation network (S), a confidence network (D), and the difficulty-aware attention mechanism. Note, we pursue *a perfect* D in this framework.

Segmentation Network

$$L_{Dice}(\mathbf{X}, \mathbf{P}; \theta_{\mathbf{S}}) = 1 - 2\frac{\sum_{c=1}^{C} \pi_c \sum_{h=1}^{H} \sum_{w=1}^{W} \sum_{t=1}^{T} P_{h,w,t,c} \widehat{P}_{h,w,t,c}}{\sum_{c=1}^{C} \pi_c \sum_{h=1}^{H} \sum_{w=1}^{W} \sum_{t=1}^{T} P_{h,w,t,c} + \widehat{P}_{h,w,t,c}}, \tag{1}$$

$$L_{CE}(X, Y; \theta_S) = -\sum_{h=1}^{H} \sum_{w=1}^{W} \sum_{t=1}^{T} \sum_{c=1}^{C} I\{Y_{h,w,t,c}\} \log \widehat{P}_{h,w,t,c} \tag{2}$$

$$L_{Hyb} = L_{Dice} + L_{CE}$$

Confidence Network

$$L_D(\mathbf{X}, \mathbf{P}; \theta_{\mathbf{D}}) = L_{BCE}(D(\mathbf{P}, \theta_{\mathbf{D}}), \mathbf{1}) + L_{BCE}(D(S(\mathbf{X}), \theta_{\mathbf{D}}), \mathbf{0}), \tag{4}$$

where

$$L_{BCE}\left(\widehat{\mathbf{Q}}, \mathbf{Q}\right) = -\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{t=1}^{T} Q_{h,w,t} \log\left(\widehat{Q}_{h,w,t}\right) + (1 - Q_{h,w,t}) \log\left(1 - \widehat{Q}_{h,w,t}\right) \tag{5}$$

$$L_{ADV}\left(\mathbf{X}, \theta_{\mathbf{S}}\right) = L_{BCE}\left(D\left(S\left(\mathbf{X}; \theta_{\mathbf{S}}\right)\right), \mathbf{1}\right) \tag{6}$$

Difficulty-aware Attention

$$L_{FDice} = 1 - 2\frac{\sum_{c=1}^{C} \pi_c (1 - dsc_c)^r \sum_{h=1}^{H} \sum_{w=1}^{W} \sum_{t=1}^{T} P_{h,w,t,c} \widehat{P}_{h,w,t,c}}{\sum_{c=1}^{C} \pi_c (1 - dsc_c)^r \sum_{h=1}^{H} \sum_{w=1}^{W} \sum_{t=1}^{T} P_{h,w,t,c} + \widehat{P}_{h,w,t,c}} \tag{7}$$

$$L_{FCE}(X, Y; \theta_S) = -\sum_{h=1}^{H} \sum_{w=1}^{W} \sum_{t=1}^{T} \sum_{c=1}^{C} I\{Y_{h,w,t}, c\} F_{h,w,t} \log \widehat{P}_{h,w,t,c} \tag{8}$$

where

$$F = (1 - M)^{\beta} \tag{9}$$

$$L_{DamHyb} = L_{FDice} + L_{FCE} \tag{10}$$

$$L_{Seg} = L_{DamHyb} + \lambda_1 L_{ADV} \tag{11}$$

实验

Table 1: DSC and ASD on the pelvic dataset by different methods.

Method	DSC (%)			ASD (in mm)		
	Bladder	Prostate	Rectum	Bladder	Prostate	Rectum
MALF	86.69(6.81)	79.28(8.72)	76.43(11.88)	1.641(.360)	2.791(.930)	3.210(2.112)
SSAE	91.75(3.10)	87.07(4.24)	86.38(4.41)	1.089(.231)	1.660(.490)	1.701(.412)
UNet	89.57(2.83)	82.22(5.88)	81.04(5.31)	1.214(.216)	1.917(.645)	2.186(0.850)
VNet	92.61(1.84)	86.40(3.61)	83.16(4.12)	1.023(.186)	1.725(.457)	1.969(.449)
DSResUNet	94.43(.90)	88.24(2.01)	86.91(3.24)	.914(.168)	1.586(.358)	1.586(.405)
Proposed	**97.48(.65)**	**92.11(1.70)**	**91.05(2.47)**	**.850(.146)**	**1.297(.276)**	**1.387(.346)**

实验

Table 2: Quantitative comparison between our proposed method and other methods on the prostate challenge testing dataset.

Method	DSC (%)			ASD (in mm)			95HD			aRVD			Score(std)
	whole	base	apex	whole	base	apex	whole	base	apex	whole	base	apex	
pxl_mcg	91.23	89.07	88.54	1.60	1.76	1.57	4.47	4.48	3.64	2.08	-0.07	2.23	88.98(3.41)
Isensee	91.61	90.29	88.05	1.52	1.65	1.64	4.21	4.20	3.85	3.42	1.86	3.48	88.84(2.94)
whu_mlgroup(2)	91.42	89.41	88.51	1.54	1.79	1.57	4.21	4.88	3.82	5.27	4.00	6.43	88.72(4.36)
Proposed	90.12	88.95	87.71	1.84	1.73	1.68	5.36	4.43	3.99	4.99	2.19	6.65	88.28(3.02)
tbrosch	90.46	88.51	85.29	1.70	1.91	1.90	4.91	5.04	4.57	2.14	7.22	-4.93	87.24(4.46)
whu_mlgroup(1)	90.26	89.15	88.36	1.86	1.79	1.62	5.57	4.83	3.90	9.74	10.73	9.64	87.04(5.79)
AutoDenseSeg	90.14	88.09	86.79	1.83	1.94	1.79	5.36	5.13	4.32	4.53	5.19	2.05	87.19(4.25)
CUMED	89.43	86.42	86.81	1.95	2.13	1.74	5.54	5.41	4.29	6.95	11.04	15.18	86.65(4.42)
SCIRESU	90.24	88.98	83.30	1.74	1.81	2.11	4.93	4.51	5.34	6.01	8.18	-7.33	86.41 (3.49)
QUILL(M2)	88.81	87.39	85.46	1.97	2.01	1.91	5.29	5.07	4.35	6.97	4.76	5.85	85.93(4.97)

实验

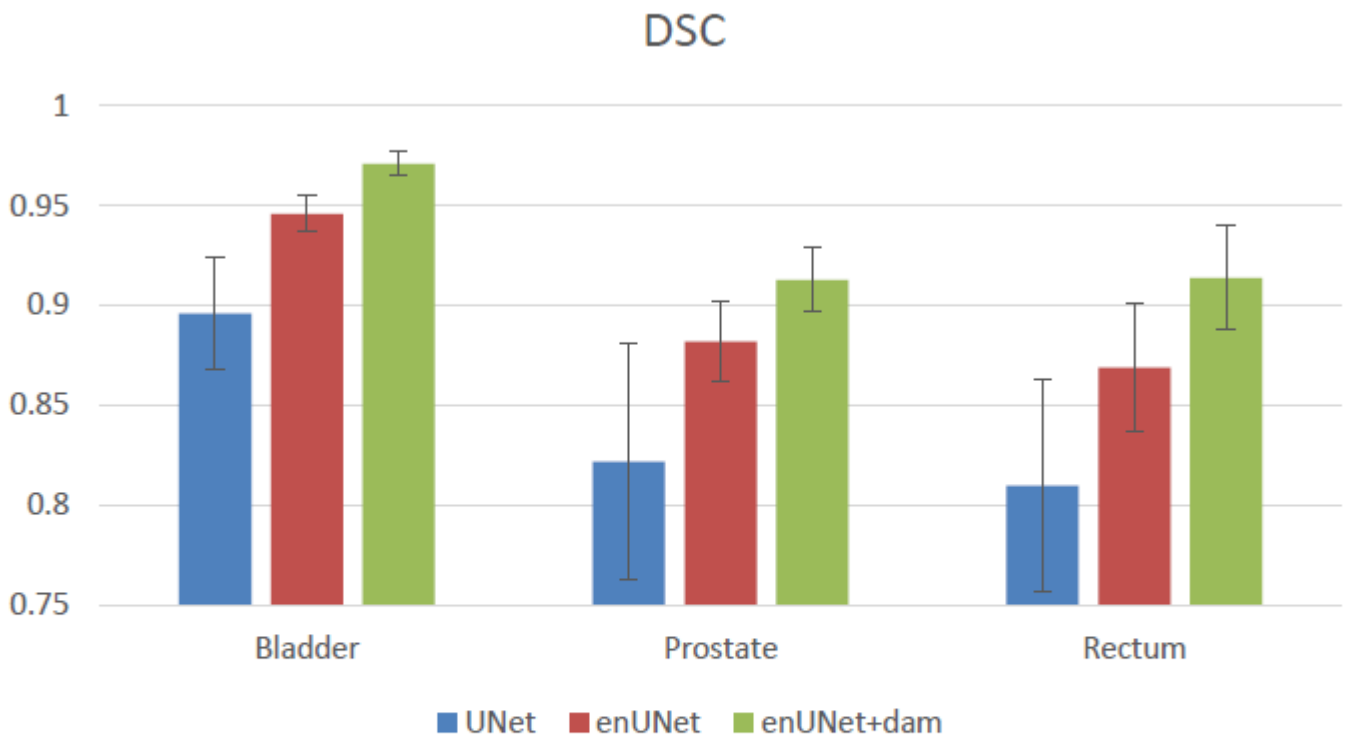

Figure 3: Average Dice ratios of different methods.

实验

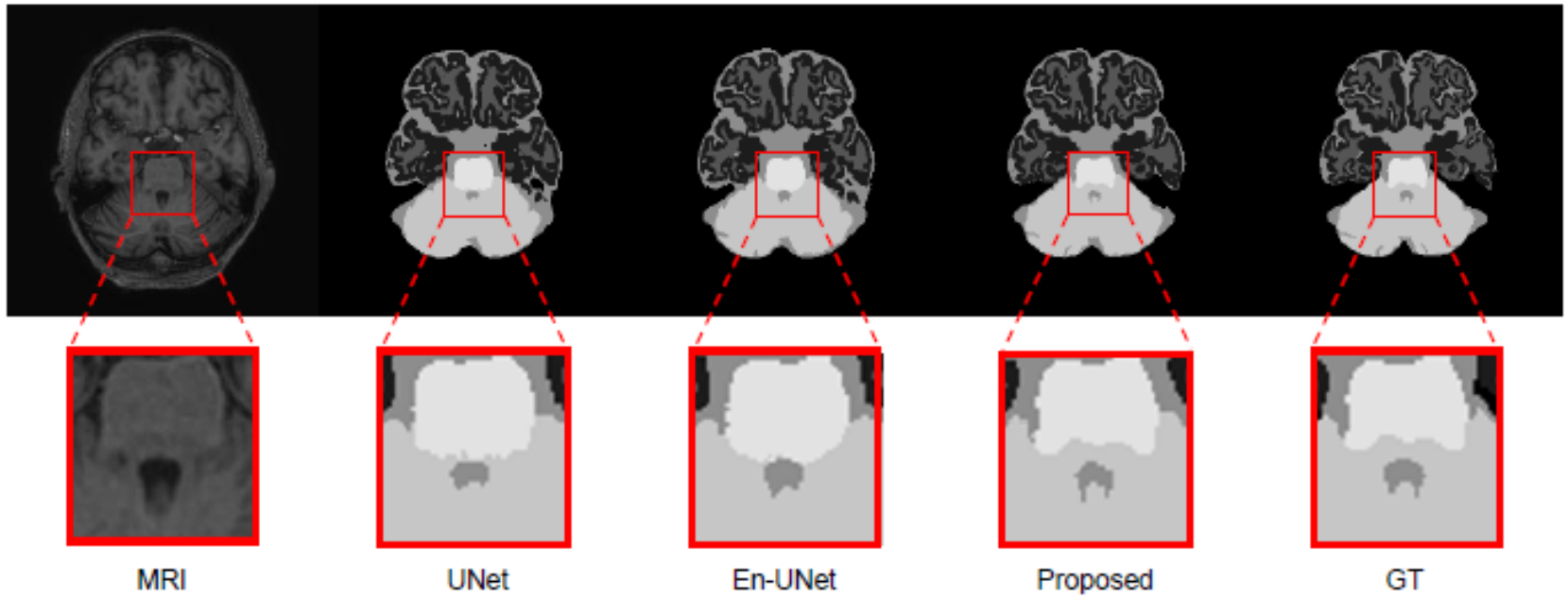

Figure 5: Comparison of segmentation results with different methods and the manual ground-truth on a sample subject.

总结

- Confidence Network
- Difficulty-aware attention mechanism was proposed to effectively address the easy-to segment sample dominance issue in a more structured way.

Thank You!